

SQL: DQL

SELECT

```
SELECT [DISTINCT] lista_colonne  
FROM tabella1  
WHERE condizione  
ORDER BY colonna [ASC | DESC]  
LIMIT n;
```





SELECT-FROM

- Elenco degli Stati e relativa popolazione

country	region	population	area
Afghanistan	ASIA	31.056.997	647.500
Albania	EUROPE	3.581.655	28.748
Algeria	AFRICA	32.930.091	2.381.740
American Samoa	OCEANIA	57.794	199
Andorra	EUROPE	71.201	468
Angola	AFRICA	12.127.071	1.246.700
Anguilla	AMERICA	13.477	102
...	...	...	...



SELECT country, population
FROM countries;

country	population
Afghanistan	31.056.997
Albania	3.581.655
Algeria	32.930.091
American Samoa	57.794
Andorra	71.201
Angola	12.127.071
Anguilla	13.477
...	...



SELECT DISTINCT

- Quali sono i continenti del mondo?

country	region	population	area
Afghanistan	ASIA	31.056.997	647.500
...	...	...	...

SELECT region
FROM countries;



region
ASIA
EUROPE
AFRICA
OCEANIA
EUROPE
AFRICA
AMERICA
AMERICA
AMERICA
...

SELECT DISTINCT region
FROM countries;



region
ASIA
EUROPE
AFRICA
OCEANIA
AMERICA

SELECT-FROM-WHERE

- Quali Stati hanno una superficie superiore a 2.000.000 mq?

```
SELECT country, area
FROM countries
WHERE area>2000000;
```

country	region	population	area
Afghanistan	ASIA	31.056.997	647.500
Albania	EUROPE	3.581.655	28.748
Algeria	AFRICA	32.930.091	2.381.740
...	...	...	...



country	area
Algeria	2.381.740
Argentina	2.766.890
Australia	7.686.850
Brazil	8.511.965
Canada	9.984.670
China	9.596.960
Congo, Dem. Rep.	2.345.410
Greenland	2.166.086
India	3.287.590
Kazakhstan	2.717.300
Russia	17.075.200
Sudan	2.505.810
United States	9.631.420

SELECT-FROM-WHERE

► Quali Stati sono in Europa?

```
SELECT country
FROM countries
WHERE region='EUROPE';
```

country	region	population	area
Afghanistan	ASIA	31.056.997	647.500
Albania	EUROPE	3.581.655	28.748
Algeria	AFRICA	32.930.091	2.381.740
...	...	...	...



country
Albania
Andorra
Austria
Belarus
Belgium
...

SELECT-FROM-WHERE

- Quali Stati europei hanno una popolazione inferiore a 50.000?

```
SELECT country, population
FROM countries
WHERE region='EUROPE' AND
      population<50000;
```

country	region	population	area
Afghanistan	ASIA	31.056.997	647.500
Albania	EUROPE	3.581.655	28.748
Algeria	AFRICA	32.930.091	2.381.740
...	...	...	...



country	population
Faroe Islands	47.246
Gibraltar	27.928
Liechtenstein	33.987
Monaco	32.543
San Marino	29.251



ORDER BY

- Stati asiatici ordinati in ordine **crescente** di popolazione.

```
SELECT *  
FROM countries  
WHERE region='ASIA'  
ORDER BY population;
```

country	region	population	area
Afghanistan	ASIA	31.056.997	647.500
Albania	EUROPE	3.581.655	28.748
Algeria	AFRICA	32.930.091	2.381.740
...	...	...	...



country	region	population	area
Maldives	ASIA	359.008	300
Brunei	ASIA	379.444	5.770
Macau	ASIA	453.125	28
Bahrain	ASIA	698.585	665
Cyprus	ASIA	784.301	9.250
Qatar	ASIA	885.359	11.437
East Timor	ASIA	1.062.777	15.007
...	...	...	...



ORDER BY

- Stati asiatici ordinati in ordine **decrescente** di popolazione.

```
SELECT *  
FROM countries  
WHERE region='ASIA'  
ORDER BY population DESC;
```

country	region	population	area
Afghanistan	ASIA	31.056.997	647.500
Albania	EUROPE	3.581.655	28.748
Algeria	AFRICA	32.930.091	2.381.740
...	...	...	...



country	region	population	area
China	ASIA	1.313.973.713	9.596.960
India	ASIA	1.095.351.995	3.287.590
Indonesia	ASIA	245.452.739	1.919.440
Pakistan	ASIA	165.803.560	803.940
Bangladesh	ASIA	147.365.352	144.000
Japan	ASIA	127.463.611	377.835
Philippines	ASIA	89.468.677	300.000
...	...	...	...

LIMIT

- I tre Stati asiatici più popolati.

```
SELECT *  
FROM countries  
WHERE region='ASIA'  
ORDER BY population DESC  
LIMIT 3;
```

country	region	population	area
Afghanistan	ASIA	31.056.997	647.500
Albania	EUROPE	3.581.655	28.748
Algeria	AFRICA	32.930.091	2.381.740
...	...	...	...



country	region	population	area
China	ASIA	1.313.973.713	9.596.960
India	ASIA	1.095.351.995	3.287.590
Indonesia	ASIA	245.452.739	1.919.440

SELECT

```
SELECT [DISTINCT] lista_colonne  
FROM tabella1  
WHERE condizione  
GROUP BY lista_colonne  
HAVING condizione_sui_gruppi  
ORDER BY colonna [ASC | DESC]  
LIMIT n OFFSET m;
```





- **COUNT, SUM, AVG, MIN e MAX** sono dette funzioni di aggregazione perché elaborano i dati contenuti in una tabella e restituiscono un singolo valore.
 - Non includono nel calcolo i valori di tipo NULL presenti nella colonna specificata come argomento.
 - È possibile utilizzare COUNT(*) per ottenere il numero di righe di una tabella, conteggiate anche se contenenti valori NULL



SUM

- Qual è la popolazione europea?

```
SELECT SUM(population)
FROM countries
WHERE region='EUROPE';
```

country	region	population	area
Afghanistan	ASIA	31.056.997	647.500
Albania	EUROPE	3.581.655	28.748
Algeria	AFRICA	32.930.091	2.381.740
...	...	...	...



SUM(population)
727.803.762



ALIAS

- Qual è la popolazione europea?
- Risultato con intestazione della colonna migliore...

country	region	population	area
Afghanistan	ASIA	31.056.997	647.500
Albania	EUROPE	3.581.655	28.748
Algeria	AFRICA	32.930.091	2.381.740
...	...	...	...



Total population
727.803.762

```
SELECT SUM(population) AS 'Total population'  
FROM countries  
WHERE region='EUROPE';
```



COUNT

- Quanti sono gli Stati africani?

country	region	population	area
Afghanistan	ASIA	31.056.997	647.500
Albania	EUROPE	3.581.655	28.748
Algeria	AFRICA	32.930.091	2.381.740
...	...	...	...

```
SELECT COUNT(*) AS 'African countries'  
FROM countries  
WHERE region='AFRICA';
```



African countries
54



GROUP BY

- Quanti sono gli Stati per ciascun continente?

country	region	population	area
Afghanistan	ASIA	31.056.997	647.500
Albania	EUROPE	3.581.655	28.748
Algeria	AFRICA	32.930.091	2.381.740
...	...	...	...



```
SELECT region, COUNT(*) AS N  
FROM countries  
GROUP BY region;
```

region	N
ASIA	52
EUROPE	47
AFRICA	54
OCEANIA	21
AMERICA	50

GROUP BY

- Quanti sono gli Stati per ciascun continente?

country	region	population	area
Afghanistan	ASIA	31.056.997	647.500
Albania	EUROPE	3.581.655	28.748
Algeria	AFRICA	32.930.091	2.381.740
American Samoa	OCEANIA	57.794	199
Andorra	EUROPE	71.201	468
Angola	AFRICA	12.127.071	1.246.700
Anguilla	AMERICA	13.477	102
Antigua & Barbuda	AMERICA	69.108	443
Argentina	AMERICA	39.921.833	2.766.890
Armenia	ASIA	2.976.372	29.800
Aruba	AMERICA	71.891	193
Australia	OCEANIA	20.264.082	7.686.850
Austria	EUROPE	8.192.880	83.870
Azerbaijan	ASIA	7.961.619	86.600
Bahamas, The	AMERICA	303.770	13.940
...	...	...	...



FROM countries
GROUP BY region;

country	region	population	area
Algeria	AFRICA	32.930.091	2.381.740
Angola	AFRICA	12.127.071	1.246.700
...	...	...	...
Anguilla	AMERICA	13.477	102
Antigua & Barbuda	AMERICA	69.108	443
...	...	...	...
Armenia	ASIA	2.976.372	29.800
Azerbaijan	ASIA	7.961.619	86.600
...	...	...	...
Albania	EUROPE	3.581.655	28.748
Andorra	EUROPE	71.201	468
...	...	...	...
American Samoa	OCEANIA	57.794	199
Australia	OCEANIA	20.264.082	7.686.850
...	...	...	...



GROUP BY

- Quanti sono gli Stati per ciascun continente?

country	region	population	area
Algeria	AFRICA	32.930.091	2.381.740
Angola	AFRICA	12.127.071	1.246.700
...	...	...	...
Anguilla	AMERICA	13.477	102
Antigua & Barbuda	AMERICA	69.108	443
...	...	...	...
Armenia	ASIA	2.976.372	29.800
Azerbaijan	ASIA	7.961.619	86.600
...	...	...	...
Albania	EUROPE	3.581.655	28.748
Andorra	EUROPE	71.201	468
...	...	...	...
American Samoa	OCEANIA	57.794	199
Australia	OCEANIA	20.264.082	7.686.850
...	...	...	...



```
SELECT *  
FROM countries  
GROUP BY region;
```

country	region	population	area
Algeria	AFRICA	32.930.091	2.381.740
Anguilla	AMERICA	13.477	102
Armenia	ASIA	2.976.372	29.800
Albania	EUROPE	3.581.655	28.748
American Samoa	OCEANIA	57.794	199



GROUP BY

- Quanti sono gli Stati per ciascun continente?

```
SELECT region, COUNT(*) AS 'N'  
FROM countries  
GROUP BY region;
```

country	region	population	area
Algeria	AFRICA	32.930.091	2.381.740
Anguilla	AMERICA	13.477	102
Armenia	ASIA	2.976.372	29.800
Albania	EUROPE	3.581.655	28.748
American Samoa	OCEANIA	57.794	199



region	N
AFRICA	52
AMERICA	47
ASIA	54
EUROPE	21
OCEANIA	50

GROUP BY + HAVING

- Quali continenti hanno almeno 50 nazioni?

```
SELECT region, COUNT(*) AS N
FROM countries
GROUP BY region
HAVING N >= 50;
```

country	region	population	area
Afghanistan	ASIA	31.056.997	647.500
Albania	EUROPE	3.581.655	28.748
Algeria	AFRICA	32.930.091	2.381.740
...	...	...	...



region	N
ASIA	52
EUROPE	47
AFRICA	54
OCEANIA	21
AMERICA	50

WHERE + GROUP BY + HAVING

- Quali continenti hanno almeno 15 nazioni di superficie maggiore di 300.000 kmq?

country	region	population	area
Afghanistan	ASIA	31.056.997	647.500
Albania	EUROPE	3.581.655	28.748
Algeria	AFRICA	32.930.091	2.381.740
...	...	...	...



region	#
ASIA	21
AFRICA	27

WHERE + GROUP BY + HAVING

- Quali continenti hanno almeno 15 nazioni di superficie maggiore di 300.000 kmq?

country	region	population	area
Afghanistan	ASIA	31.056.997	647.500
Albania	EUROPE	3.581.655	28.748
Algeria	AFRICA	32.930.091	2.381.740
...	...	...	...

```
SELECT *  
FROM countries  
WHERE area>300000
```

WHERE + GROUP BY + HAVING

- Quali continenti hanno almeno 15 nazioni di superficie maggiore di 300.000 kmq?

```
SELECT region, COUNT(*) AS N
FROM countries
WHERE area>300000
GROUP BY region
```

country	region	population	area
Afghanistan	ASIA	31.056.997	647.500
Albania	EUROPE	3.581.655	28.748
Algeria	AFRICA	32.930.091	2.381.740
...	...	...	...



region	N
ASIA	21
EUROPE	11
AFRICA	27
OCEANIA	2
AMERICA	11

WHERE + GROUP BY + HAVING

- Quali continenti hanno almeno 15 nazioni di superficie maggiore di 300.000 kmq?

```
SELECT region, COUNT(*) AS N
FROM countries
WHERE area > 300000
GROUP BY region
HAVING N >= 15;
```

country	region	population	area
Afghanistan	ASIA	31.056.997	647.500
Albania	EUROPE	3.581.655	28.748
Algeria	AFRICA	32.930.091	2.381.740
...	...	...	...



region	N
ASIA	21
EUROPE	11
AFRICA	27
OCEANIA	2
AMERICA	11



region	N
ASIA	21
AFRICA	27



Esercizi

- Qual è la popolazione media degli Stati europei?
- Qual è la popolazione totale in ciascun continente?

SELECT

```
SELECT [DISTINCT] lista_colonne  
FROM tabella1  
    [INNER JOIN tabella2 ON condizione]  
    [LEFT JOIN tabella3 ON condizione]  
    [RIGHT JOIN tabella4 ON condizione]  
WHERE condizione  
GROUP BY lista_colonne  
HAVING condizione_sui_gruppi  
ORDER BY colonna [ASC | DESC]  
LIMIT n OFFSET m;
```



INNER JOIN

```
SELECT *  
FROM orders INNER JOIN customers;
```

orders		
<u>orderID</u>	customerID	total
1	10	15,5
2	11	30
3	10	49,99

customers		
<u>customerID</u>	firstName	lastName
10	Andrea	Pisani
11	Luca	Peracino



Prodotto cartesiano					
<u>orderID</u>	<u>customerID</u>	<u>total</u>	<u>customerID</u>	firstName	lastName
1	10	15,5	10	Andrea	Pisani
1	10	15,5	11	Luca	Peracino
2	11	30	10	Andrea	Pisani
2	11	30	11	Luca	Peracino
3	10	49,99	10	Andrea	Pisani
3	10	49,99	11	Luca	Peracino

INNER JOIN

```
SELECT *  
FROM orders AS o INNER JOIN customers AS c  
ON o.customerID = c.customerID;
```

orders		
orderID	customerID	total
1	10	15,5
2	11	30
3	10	49,99

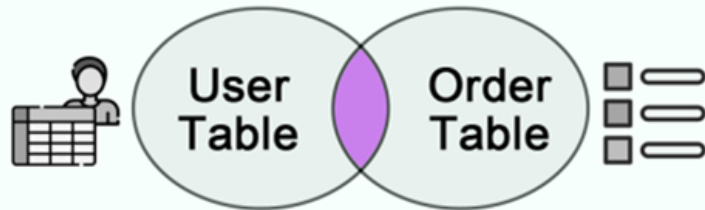
customers		
customerID	firstName	lastName
10	Andrea	Pisani
11	Luca	Peracino



Prodotto cartesiano					
orderID	customerID	total	customerID	firstName	lastName
1	10	15,5	10	Andrea	Pisani
1	10	15,5	11	Luca	Peracino
2	11	30	10	Andrea	Pisani
2	11	30	11	Luca	Peracino
3	10	49,99	10	Andrea	Pisani
3	10	49,99	11	Luca	Peracino

LEFT JOIN

INNER JOIN



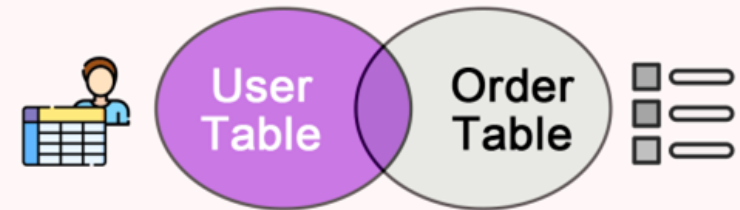
User ID	User Name
123	Bob
124	Alice
125	Carrie

User ID	Order ID
123	333
123	222
126	111

SELECT*
FROM USER_TABLE a
INNER JOIN ORDER_TABLE b
ON a.USER_ID = b.USER_ID

User ID	User Name	Order ID
123	Bob	333
123	Bob	222

LEFT JOIN



User ID	User Name
123	Bob
124	Alice
125	Carrie

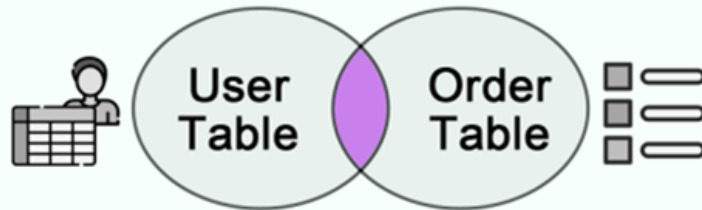
User ID	Order ID
123	333
124	111
126	666

SELECT*
FROM USER_TABLE a
LEFT JOIN ORDER_TABLE b
ON a.USER_ID = b.USER_ID

User ID	User Name	Order ID
123	Bob	333
124	Alice	111
125	Carrie	NULL

RIGHT JOIN

INNER JOIN



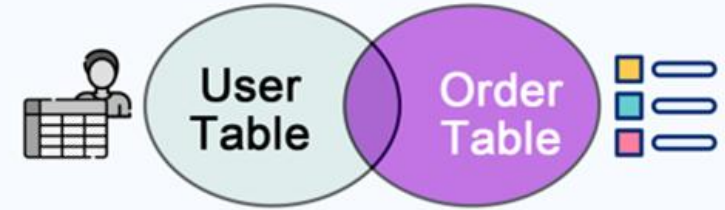
User ID	User Name
123	Bob
124	Alice
125	Carrie

User ID	Order ID
123	333
123	222
126	111

SELECT*
FROM USER_TABLE a
INNER JOIN ORDER_TABLE b
ON a.USER_ID = b.USER_ID

User ID	User Name	Order ID
123	Bob	333
123	Bob	222

RIGHT JOIN



User ID	User Name
123	Bob
124	Alice
125	Carrie

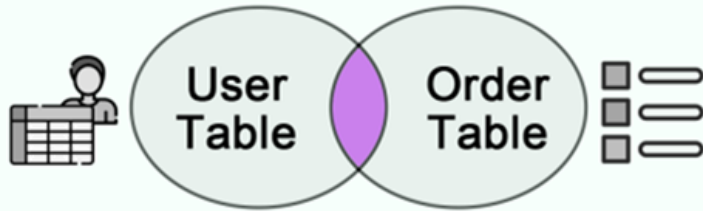
User ID	Order ID
123	333
124	111
126	666

SELECT*
FROM USER_TABLE a
RIGHT JOIN ORDER_TABLE b
ON a.USER_ID = b.USER_ID

User ID	User Name	Order ID
123	Bob	333
124	Alice	111
126	NULL	666

FULL OUTER JOIN

INNER JOIN



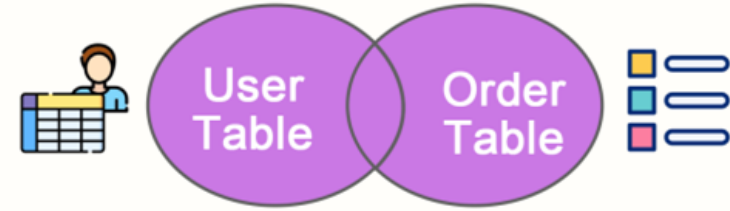
User ID	User Name
123	Bob
124	Alice
125	Carrie

User ID	Order ID
123	333
123	222
126	111

```
SELECT*  
FROM USER_TABLE a  
INNER JOIN ORDER_TABLE b  
ON a.USER_ID = b.USER_ID
```

User ID	User Name	Order ID
123	Bob	333
123	Bob	222

FULL OUTER JOIN



User ID	User Name
123	Bob
124	Alice
125	Carrie

User ID	Order ID
123	333
124	111
126	666

```
SELECT*  
FROM USER_TABLE a  
FULL OUTER JOIN ORDER_TABLE b  
ON a.USER_ID = b.USER_ID
```

User ID	User Name	Order ID
123	Bob	333
124	Alice	111
125	Carrie	NULL
126	NULL	666